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Paper 1: Tutorial on Causal Mediation Analysis for Pharmacometricians

Bibliographic Information

- **Title:** Tutorial on Causal Mediation Analysis for Pharmacometricians
- **Authors:** N/A
- **Journal:** N/A
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Paper Type: GENERIC

Quick Take

This tutorial introduces causal mediation analysis for pharmacometricians, using counterfactual definitions and a simulation-based workflow. It demonstrates how to decompose total treatment effects into direct and indirect effects mediated by a biomarker, with applications to PK-PD modeling. The paper provides practical guidance on implementation, assumptions, and interpretation.

Executive Summary

This tutorial provides a comprehensive introduction to causal mediation analysis tailored for pharmacometricians. It begins with counterfactual definitions of total, direct, and indirect effects, distinguishing between pure indirect effect (PIE) and total indirect effect (TIE). A general simulation-based workflow is presented, leveraging Monte Carlo simulations from joint PK-PD models of mediator and outcome. Two case examples illustrate the approach: a simple linear example and a more complex nonlinear mixed-effects model with disease subtypes, dose levels, and time-varying effects. The tutorial discusses critical assumptions (no unmeasured confounding, correct model specification), interpretation of proportion mediated, and practical considerations such as choice of treatment, mediator, and outcome variables. It emphasizes that causal mediation analysis is a natural fit for pharmacometrics due to its flexibility with nonlinearities and interactions, and provides example code and datasets.

Scientific Context & Motivation

Biomarkers are crucial for studying pharmacodynamics and efficacy, and may serve as surrogate endpoints if validated. Mediation analysis quantifies the proportion of a treatment effect on a clinical outcome that is mediated by a biomarker, supporting biomarker development and regulatory decisions. Traditional mediation methods (e.g., difference method) are limited with nonlinear models or interactions. Causal mediation analysis, based on counterfactual reasoning, offers general definitions that apply to complex PK-PD models. This tutorial addresses the gap between causal inference theory and pharmacometric practice, providing a practical workflow.

✂ Methodological Snapshot

The tutorial presents a four-step simulation-based workflow for causal mediation analysis using nonlinear mixed-effects models. Step 1: Fit a joint PK-PD model for mediator M and outcome Y to observed data. Step 2: Sample J sets of model parameters from the estimated uncertainty distribution (e.g., multivariate normal from covariance matrix). Step 3: For each parameter set, perform Monte Carlo simulation of N individuals (e.g., $N = 10,000$) by: (a) simulating mediator values under all treatment conditions of interest (e.g., $M(0)$, $M(dose1)$, $M(dose2)$); (b) simulating counterfactual outcomes for combinations of treatment and mediator (e.g., $Y(0, M(0))$, $Y(0, M(dose))$, $Y(dose, M(0))$, $Y(dose, M(dose))$); (c) computing individual-level causal effects (TE, PIE, TIE) and averaging to obtain population effects. Step 4: Summarize across J samples to obtain point estimates and confidence intervals (e.g., percentiles). The workflow handles nonlinearities, interactions, covariates, and random effects naturally. Residual error can be included or excluded depending on whether the mediator’s residual variation is considered predictive of outcome.

📋 Detailed Analysis

The tutorial excels in bridging causal inference theory with pharmacometric practice. The counterfactual definitions are clearly explained with simple examples, and the distinction between PIE and TIE is well-motivated. The simulation-based workflow is a strength because it avoids model-specific analytical derivations and leverages pharmacometricians’ familiarity with Monte Carlo simulation. The case examples effectively illustrate how mediation can vary by dose, time, and covariates, and how model structure (e.g., interactions, effect compartments) influences the results. The discussion of assumptions is thorough, though the tutorial could have emphasized more strongly the need for sensitivity analyses in practice. The interpretation of proportion mediated is nuanced, especially when direct and indirect effects oppose. The tutorial also touches on practical issues like choice of treatment variable (dose vs. exposure) and mediator (predicted vs. observed). One limitation is that the workflow assumes the PK-PD model is correctly specified; model misspecification could lead to biased mediation estimates. Additionally, the computational burden of the simulation-based approach may be high for very complex models, though this is not discussed. Overall, this is a valuable resource for pharmacometricians looking to incorporate mediation analysis into their work.

Domain Context

Causal mediation analysis has been widely used in epidemiology and social sciences, but its adoption in pharmacometrics has been limited. This tutorial addresses that gap by providing a practical framework that aligns with standard pharmacometric activities: longitudinal modeling of biomarkers and outcomes, simulation, and covariate analysis. The approach is particularly relevant for biomarker validation and surrogate endpoint evaluation in drug development. The tutorial also connects to regulatory concepts (e.g., Fleming’s four-category system for efficacy measures) and highlights how mediation analysis can support dosing decisions and bridging studies. The use of joint PK-PD models to account for correlations between mediator and outcome

random effects is a pharmacometric-specific contribution that strengthens the causal interpretation.

☒ Key Findings

The paper presents a simulation-based causal mediation analysis workflow for pharmacometricians. Key concepts include: (1) counterfactual definitions of total effect (TE), pure indirect effect (PIE), and total indirect effect (TIE); (2) proportion mediated (PM) as IE/TE ; (3) a four-step workflow: fit joint PK-PD models, sample from parameter uncertainty, perform Monte Carlo simulations for counterfactual scenarios, and compute causal effects; (4) application to a complex PK-PD model showing how mediation varies by dose, time, and disease subtype; (5) discussion of assumptions (no unmeasured confounding, correct model) and interpretation of PM (can exceed 100% if direct and indirect effects oppose). The tutorial emphasizes that causal mediation analysis is flexible and suitable for nonlinear mixed-effects models.

Strengths & Limitations

Strengths

- Provides a clear, step-by-step tutorial with example code and datasets, making it accessible to pharmacometricians.
- Uses a general simulation-based approach that works with any PK-PD model, including nonlinearities and interactions.
- Distinguishes between pure indirect effect (PIE) and total indirect effect (TIE), clarifying their interpretation.
- Illustrates the workflow with two case examples, including a realistic complex model with covariates and time-varying effects.
- Discusses practical considerations such as choice of treatment variable (dose vs. exposure), mediator (predicted vs. observed), and outcome (landmark time vs. model parameter).
- Addresses key assumptions and their implications, including the need for joint modeling to account for random effects correlations.

Limitations (Acknowledged by Authors)

- The approach relies on strong non-confounding assumptions that cannot be tested from observed data.
- Sensitivity analyses for unmeasured confounders are recommended but rarely used in practice.
- The proportion mediated can be difficult to interpret when direct and indirect effects have opposite directions.
- Extension to multiple mediators requires additional strong assumptions.

Limitations (Expert Review)

- The tutorial does not provide analytical solutions for causal effects, which may be needed for models where simulation is computationally expensive.

- The workflow assumes the PK-PD model is correctly specified; model misspecification could bias mediation estimates.
- The example uses a single simulated dataset; performance under different study designs or sample sizes is not explored.
- Guidance on selecting the number of Monte Carlo samples (N) and parameter samples (J) is heuristic; formal power analysis is absent.

Generalizability

The simulation-based workflow is general and can be applied to any PK-PD model with continuous or discrete mediator/outcome. However, generalizability of specific mediation estimates depends on the study population and design; the tutorial notes that predictions for new populations require careful consideration of covariate distributions and model applicability.

Key Equations

Total Effect (TE)

$$TE = E[Y(1) - Y(0)]$$

The total causal effect of treatment ($A=1$) versus no treatment ($A=0$) on the outcome Y .

Pure Indirect Effect (PIE)

$$PIE = E[Y(0, M(1)) - Y(0, M(0))]$$

The indirect effect mediated by the biomarker M , setting treatment to untreated but mediator to its value under treatment.

Total Indirect Effect (TIE)

$$TIE = E[Y(1, M(1)) - Y(1, M(0))]$$

The indirect effect mediated by M , setting treatment to treated but mediator to its value under no treatment.

Proportion Mediated (PM)

$$PM = \frac{IE}{TE} \times 100\%$$

The percentage of the total effect that is mediated by the biomarker. IE can be PIE or TIE.

Figures & Tables

- **Figure 1:** Schematic of the PK-PD model used in Case example 2, showing drug concentration driving mediator M and direct effect on outcome Y , with M also affecting Y . Includes effect compartments (keo1, keo2) and disease subtype covariate.
 - *Significance:* Illustrates the complex model structure with nonlinearities and interactions used to demonstrate the simulation-based mediation analysis.

- **Figure 2:** Boxplots of simulated mediator M and outcome Y data over time, stratified by dose group (placebo, 50 mg, 200 mg).
 - *Significance:* Shows the raw data used for model estimation, highlighting treatment effects and variability.
 - **Figure 3:** Causal mediation analysis results at Day 730, stratified by disease subtype and dose. Shows total effect, PIE, TIE, and proportion mediated with confidence intervals.
 - *Significance:* Demonstrates how mediation varies by dose and disease subtype, and the difference between PIE and TIE due to interaction effects.
 - **Figure 4:** Illustration of proportion mediated when direct and indirect effects have opposite directions, showing PM can exceed 100% or be negative.
 - *Significance:* Highlights the importance of interpreting PM in context, especially when direct and indirect effects oppose each other.
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Code & Reproducibility Assessment

Example code and datasets are provided as Supporting Information (R code for the simulation-based workflow using NONMEM and R).

Future Directions

Future work could extend the workflow to multiple mediators, incorporate sensitivity analyses for unmeasured confounding, and develop guidelines for study design optimization for mediation analysis. Additionally, applying the approach to real-world datasets and comparing with traditional mediation methods would strengthen practical recommendations.

Expert Commentary

This tutorial fills an important gap by translating causal mediation concepts into a practical pharmacometric framework. The simulation-based approach is particularly appealing because it aligns with standard pharmacometric workflows and can handle the complexity typical of PK-PD models. The distinction between PIE and TIE is a nuanced but critical point that is often overlooked. I would encourage pharmacometricians to adopt this methodology for biomarker evaluation, but to remain vigilant about the strong assumptions required, especially regarding unmeasured confounding. The inclusion of example code is a major strength that lowers the barrier to implementation.

Bottom Line

Causal mediation analysis is a valuable addition to the pharmacometrician's toolbox for quantifying biomarker mediation of treatment effects. The simulation-based workflow presented is flexible, intuitive, and leverages existing pharmacometric skills in nonlinear mixed-effects modeling and Monte Carlo simulation. Practitioners should carefully consider the required assumptions (especially no unmeasured confounding) and interpret proportion mediated in the context of the intended biomarker use.

☒ Figures

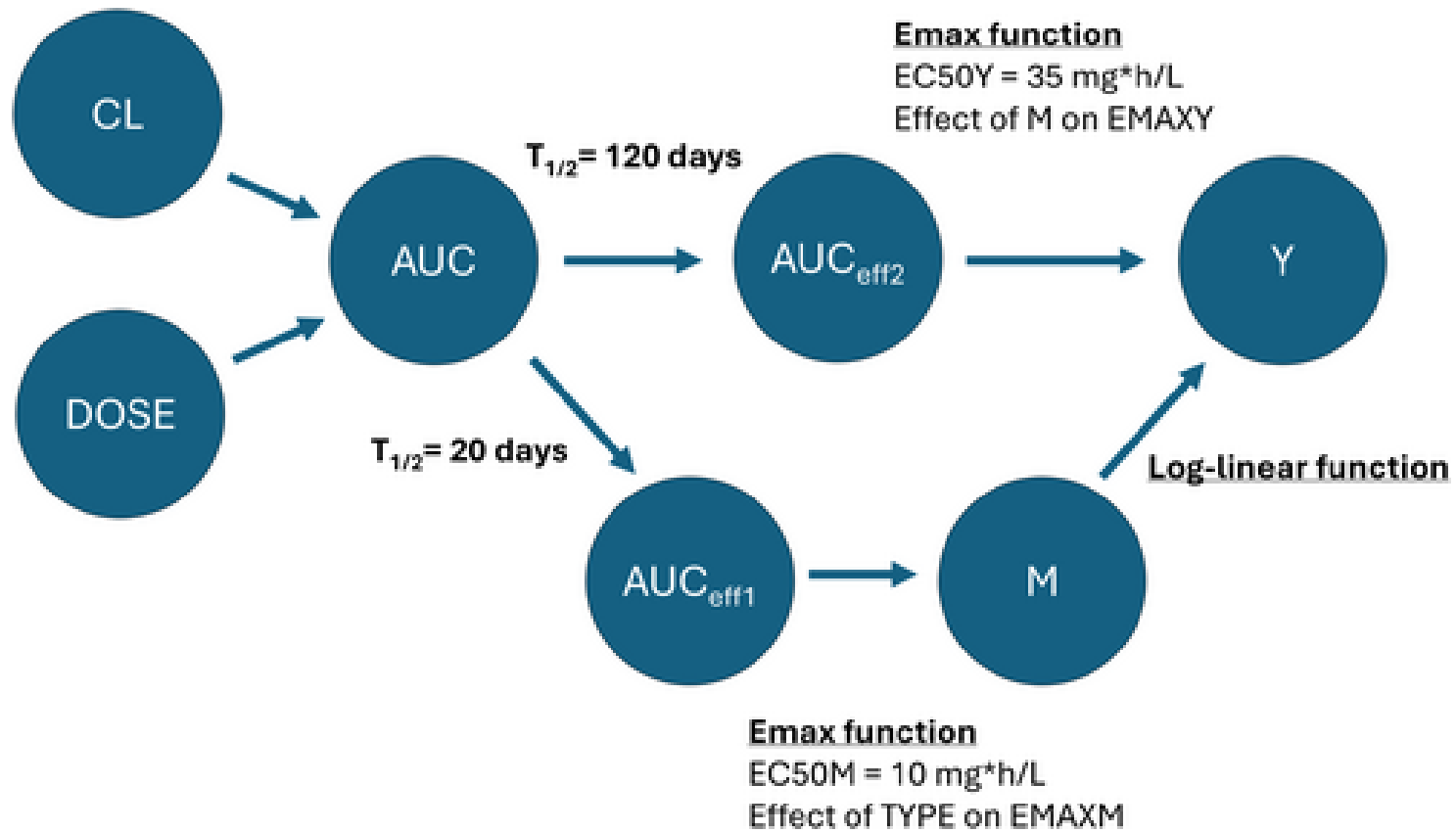


Figure 1: Schematic representation of the PK-PD model used in Case example 2 of the simulation-based causal mediation analysis workflow. In this diagram, the paths for the

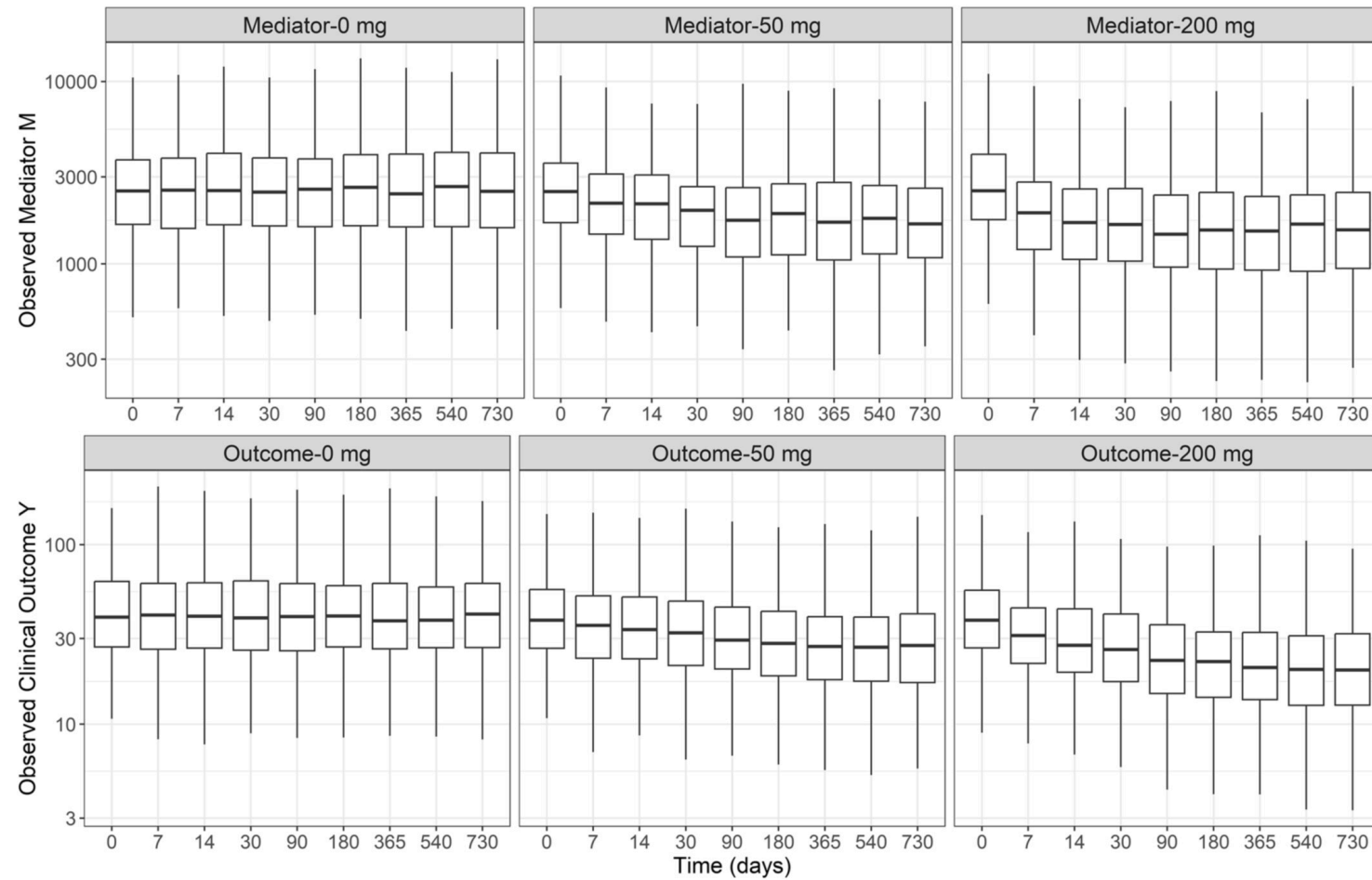


Figure 2: Overview of simulated data in Case example 2. Boxplots showing the simulated mediatorM(top) and outcomeY(bottom) data at each time point, stratified by dose group

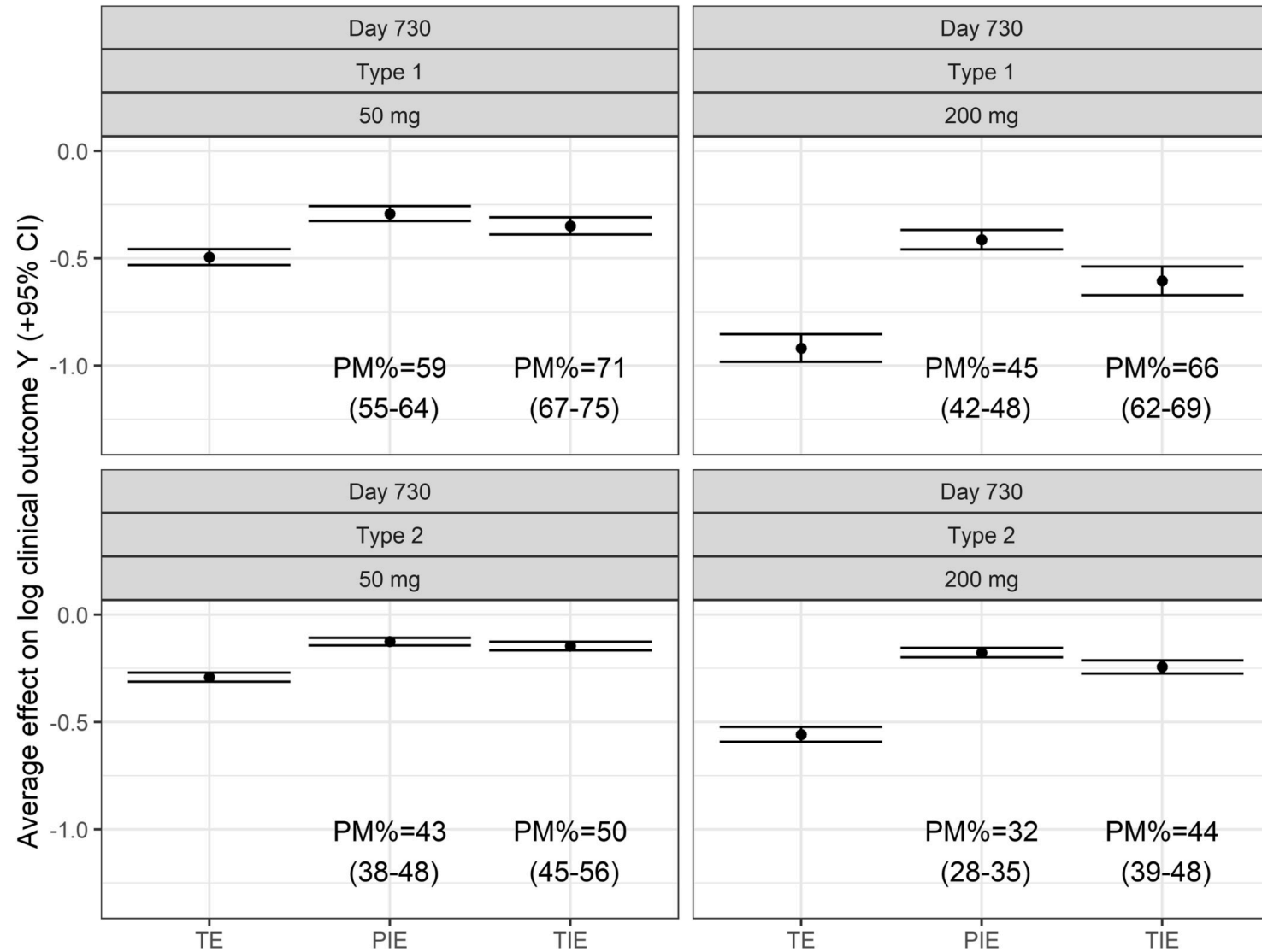


Figure 3: Causal mediation analysis of the outcome at Day 730, stratified by disease subtype (Type 1 vs. Type 2) and dose level. The average causal effects—total effect (T

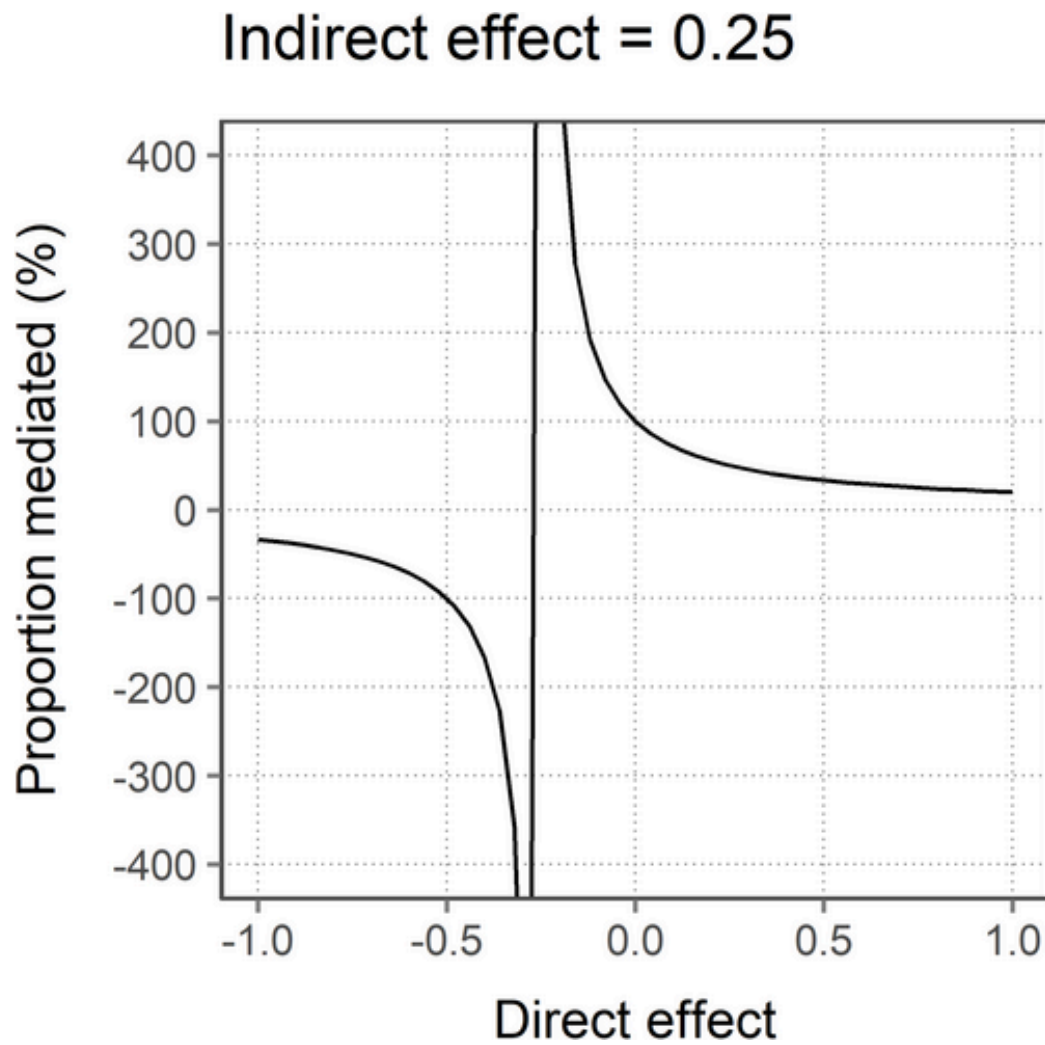


Figure 4: Illustration of the relationship between direct effect, indirect effect and proportion mediated (PM) for situation where indirect effect is constant at 0.25. When